

LESSON 1: HUMAN-COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a multidisciplinary field that focuses on the study of the design of computers and, in particular, the interaction that occurs between humans and computers.

Initially, HCI focused only on computers, but this has changed over the years to currently include almost all forms of information technologies. Understanding how humans interact with computers and other technologies has greatly changed our lives. It has made computers and technology more user-friendly and intuitive, which has greatly broadened the range of people who can comfortably use these technologies.

HUMAN-COMPUTER INTERFACE

A system's interface is the boundary across which it interacts with the outside world. The system receives input and delivers output across its interface, which is physically determined by the system's input and output devices.

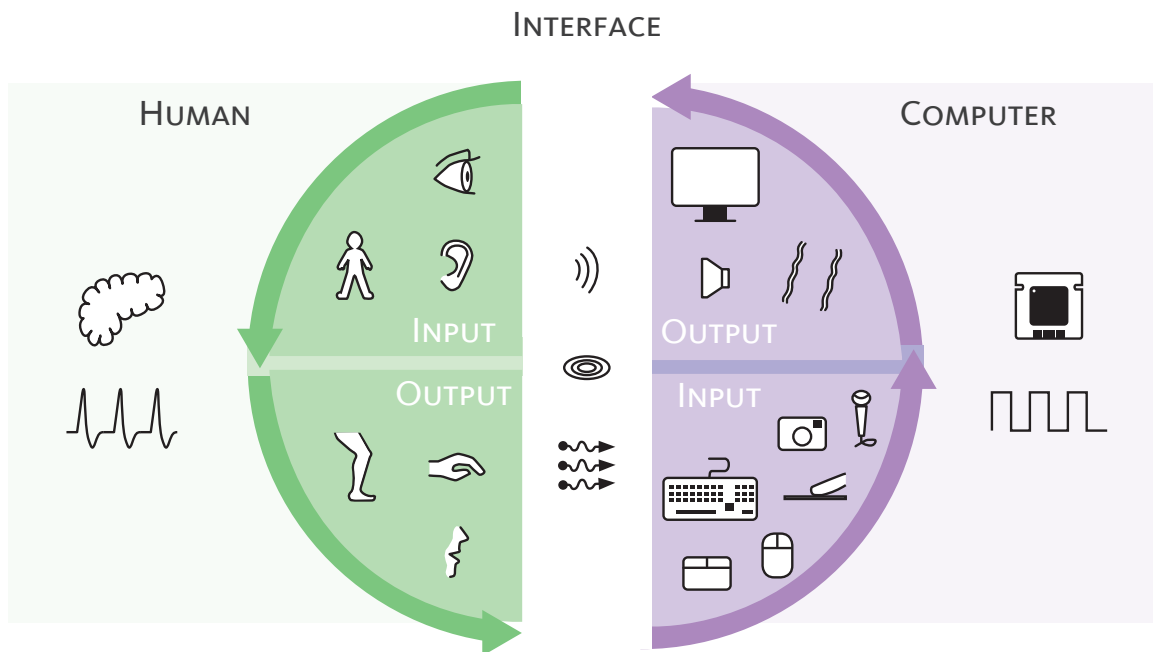
Two systems can interact if their interfaces are compatible and allow them to exchange information. Humans have designed computer interfaces to be compatible with their own. This allows humans to exchange information with computers and to interact with them. So when a human uses a computer, the boundary where the two systems meet and exchange signals is referred to as the human-computer interface.

THE HUMAN-COMPUTER INTERFACE IS THE BOUNDARY ACROSS
WHICH THE TWO SYSTEMS INTERACT.

INPUT/OUTPUT LOOP

The input/output (I/O) loop is fundamental to human-computer interaction. The interface between the two systems allows signals to be passed back and forth. The output of one system becomes the input of the other, which in turn generates an output. This process repeats itself, creating an I/O loop across the interface. The I/O loop perpetuates until one of two conditions arises: 1) the task is completed, or 2) the loop is interrupted for some reason.

HUMAN-COMPUTER INTERACTION IS A LOOP OF INPUTS AND OUTPUTS BEING EXCHANGED BY THE TWO SYSTEMS.



The human-computer interface is established by the input and output components of each system. The user receives a signal emitted from the computer. The sensory organs convert it to action potentials that the human can process and then accordingly respond. The human's response, or output, then becomes the computer's input. The incoming signal emitted by the human is converted to electronic digital signals, which the computer uses to process and respond, creating a loop of inputs and outputs.

EXAMPLE 3.1

When a user enters text into a word processor, the keyboard sends the input to the computer, which processes the input and displays the output on the screen. The user can then see the output and make changes, which are then sent back to the computer. The process repeats in a loop until the desired result is achieved, which is completing the essay, or the loop is interrupted by the user being called for dinner.

PERIPHERALS

Peripheral devices are input and output devices that can be connected to a computer, extending the human-computer interface. They augment an existing functionality or add a new one that the computer could not previously perform. Peripheral devices can act as input, output or as both input and output devices (I/O devices).

**PERIPHERAL DEVICES EXPAND THE HUMAN-COMPUTER INTERFACE
AND PROVIDE ADDITIONAL FUNCTIONALITIES.**

EXAMPLE 3.2

The audio headset is a peripheral output device. It connects to a computer and provides the user with privacy and clarity of sound. In some cases, the headset has a built-in microphone, making it an I/O device that can provide both input and output of sound. This device enhances an existing functionality by bringing the input and output devices closer to the mouth and ears of the user. This new interface provides a more direct and clear transmission of sound signals.

EXAMPLE 3.3

Another example of a peripheral device is the projector, a peripheral output device commonly used in classrooms. This technology outputs light to a much wider area than the computer screen, allowing for a much larger audience of viewers, making the interface not just with one person but several. The smartboard is another classroom peripheral device. It can both display content and allow users to manipulate it directly; it is an I/O device.

There are several peripheral devices that connect with a computer and extend its interface. Some devices connect through wired connections, while others have wireless connections.

Input Devices	Camera - Webcam - Barcode reader - Scanner - Game controller - Microphone
Output Devices	Printer - 3D Printer - Plotter - Projector - Earphones - Speakers - Television
I/O Devices	Headset - Smartboard

By extending the human-computer interface through peripheral devices, users can experience more interactions with computers, allowing them to perform a wide range of tasks across various contexts, such as work, education, entertainment and communication.

It is important to note that each of these devices has its own software, programs that contain the instructions that allow the peripheral device to exchange information with the computer. While this software may not be directly visible in the user interface, it does perform critical work behind the scenes. Without the software, the hardware will not work.

SMART MACHINES

Embedded computers have impacted our world significantly, with processors and other digital components embedded in almost every machine, ranging from home appliances to automobiles.

Embedded computers allow us to enhance the machine's interface as well as provide additional functions that can be performed using the machine. Therefore, smart machines have an interface that provides an enhanced level of functionality and usability, ensuring that the machine performs its intended tasks to the satisfaction of its user.

THE HUMAN-MACHINE INTERFACE IS THE INTERFACE ACROSS
WHICH HUMANS INTERACT WITH MACHINES.

THE SMARTPHONE

The first cellphone was released in 1973 by Motorola, the DynaTAC 8000X. It was large, weighing about 1 kg and costing over \$3,900. As a device, it was purely a phone and could only make and receive calls.

This changed in 1996 with the release of the Nokia 9000 Communicator, the first phone to have an embedded computer. This cellphone provided internet capability, with web browsing and emailing as functionalities. However, it was not until 2007 that the Apple iPhone became the first smartphone to achieve success in the market.

Nowadays, smartphones are no longer just cellphones used for the sole purpose of making calls. Instead, smartphones have become general-purpose machines capable of performing a wide range of tasks. It is now considered a personal computer and not just a smartphone.

THE CASH REGISTER

A cash register is a common example of a machine that contains an embedded computer. The human-machine interface of the register comprises:

Input Devices

Barcode scanner	It is used to scan the barcode on items. The barcode contains information like the name of the product and its price, which is input into the system.
Keyboard	It is used to input information, such as the price of items and method of payment.

Output Devices

Drawer	It is used to store cash. An actuator moves and releases the drawer, allowing it to spring forward and open.
Printer	It is used to print receipts for customers. There are two actuators for this output device: 1) a motor to turn the role of paper and 2) a printhead that moves across printing line by line.
Screen	It is used to show the transaction information, such as the price of items, total cost and change due.

Unlike personal computers that are intended to have a general purpose, the cash register is designed as a machine that serves a specific purpose. Therefore, its interface must be optimized to allow humans to perform the required tasks that serve that purpose. As such, the extent of the interaction between human and machine is limited to the tasks that need to occur at a point of sale, such as identifying items and prices, receiving payments and printing receipts.

ROBOTS

A robot is a machine capable of performing a variety of tasks autonomously or under varying degrees of control by an operator. A robot must interact with the world and humans; as such, it typically consists of input devices in the form of sensors, output devices in the form of actuators that provide movement across joints and a body that holds everything together. The design of the robot and its interface is dictated by the purpose it must serve.

ROBOTS ARE MACHINES CAPABLE OF PERFORMING TASKS AUTONOMOUSLY OR WITH SOME HUMAN INTERVENTION.

Robots can be categorized depending on their degree of autonomy, mobility and purpose.

- **Autonomy:** refers to the degree to which a robot can act independently. An autonomous robot can make decisions on its own without human intervention. A semi-autonomous robot can make most decisions on its own. In comparison, a non-autonomous robot relies on a human operator to control its actions. Therefore, the embedded computer in each of these three cases has varying degrees of decision-making capabilities that can vary from making all decisions on its own to simply assisting in coordinating input and output, leaving the decisions to the human operator.
- **Mobility:** refers to a robot's capability of moving and changing location. Some robots are stationary, such as the robotic arms used in car factories, while other robots are mobile. Mobile robots have varying degrees of mobility. Some can only negotiate a smooth, flat surface, such as robotic vacuum cleaners. In contrast others, such as some advanced humanoid robots, can walk outdoors on uneven terrain, run, jump and even do somersaults.
- **Purpose:** refers to the task or set of tasks a robot is intended to perform. Some tasks are menial and highly repetitive, such as those required in

manufacturing. While other tasks require entering difficult or dangerous locations, such as exploring underwater caves or finding people trapped by fire. Still, other robots, humanoid robots, are designed to serve the purpose of satisfying the engineering achievement of creating machines very much in our own likeness.

Robots have brought about both benefits and concerns. On the one hand, robots have taken over jobs that are unpleasant or even unsafe for humans. Menial and repetitive tasks tend to make jobs unpleasant, and robots have been designed to perform such tasks as:

1. Cleaning floors and windows.
2. Sorting and organizing items in warehouses.
3. Tightening screws or fastening components on assembly lines.
4. Packaging products in factories.

Robots have also taken over jobs and tasks that are dangerous, thereby keeping humans out of harm's way.

1. Handling hazardous materials or chemicals in environments such as nuclear power plants or chemical facilities.
2. Conducting underwater inspections or repairs in deep-sea environments.
3. Performing tasks in areas with high levels of radiation or toxic substances.
4. Exploring hostile or extreme environments like volcanoes, disaster sites and even other planets.
5. Conducting military operations, such as bomb disposal or surveillance in conflict zones.

However, robots have also raised several concerns. One concern is the displacement of humans from their jobs due to the increasing advancement and capability of robots, which can now perform tasks traditionally done by humans. According to a study by the McKinsey Global Institute, up to 800 million jobs could be lost to automation by 2030. This is equivalent to 14% of the global workforce.

Other concerns involve ethical implications. As robots become more independent, will they behave ethically and responsibly? Will they prioritize the best interests of humans or potentially cause them harm? Will they adhere to the ethical standards we hold ourselves to?

In 1942, Isaac Asimov, a science fiction author, introduced the three laws of robotics.

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Although these laws were introduced in science fiction publications, they are based on the ethical considerations that humans are concerned about.

Overall, it is important to carefully balance the benefits of robotics while addressing these concerns as we move towards a more automated future—a future in which humans interact with computers, machines, and robots.

SUMMARY

Human-Computer Interaction is a multidisciplinary field that focuses on the study of the design of computers and its impact on the interaction that occurs between humans and computers. HCI include computers and other information technologies, particularly those that have an embedded computer.

The Human-Computer Interface is the boundary across which the two systems interact. For the computer, this includes all its input and output devices. For the human, this includes the sensory organs and hands. Interaction occurs in an input/output loop that results across the interface.

The interface of the computer can be expanded by installing peripheral devices. These devices are input and output devices that augment an existing functionality or add a new one. Peripherals enhance human-computer interaction.

Embedded computers have made a wide variety of machines and electronic devices smart. Introducing embedded computers enhances the human-machine interface by adding new functionalities and improving usability.

Robots are machines capable of performing tasks autonomously or with some human intervention. They are categorized depending on their autonomy, mobility and purpose. Robots bring the benefits of taking over menial and repetitive jobs, as well as dangerous ones. The concerns are that robots are taking over too many jobs leaving people jobless, and that they may not in the future act ethically and in our best interest.

Career Tips

Mechatronics is an interdisciplinary field that combines mechanical engineering, electrical engineering, and computer science. Mechatronic systems can be found in various applications, such as manufacturing, transportation, healthcare and robotics.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Humans interact with computers across an interface.
<u>F</u>	Interaction occurs in one direction, from human to computer.
<u>T</u>	The I/O loop is a cycle of exchanges of signals.
<u>F</u>	The computer's interface cannot be expanded.
<u>T</u>	Peripheral devices are input and output devices that can expand the interface.
<u>T</u>	Embedded computers make machines smart.
<u>T</u>	The smartphone used to be a cellphone with an embedded computer; now, it is a general-purpose computer.
<u>F</u>	All robots perform tasks autonomously.
<u>T</u>	Taking over menial jobs is considered a benefit of robots.
<u>F</u>	Isaac Asimov was a scientist.

Activity 1.2

Answer the questions below.

1. Which component is not part of the interface?
 - a) **Processor**
 - b) Mouse
 - c) Keyboard

2. Which device acts as a peripheral device for a tablet?
 - a) Touchscreen
 - b) Memory
 - c) **Digital pen**

3. Which device acts as a peripheral device for a laptop?
 - a) Screen
 - b) **Printer**
 - c) Keyboard

4. Which device acts as a peripheral device for a smartphone?
 - a) **Wireless earphones**
 - b) Touchscreen
 - c) Processor

5. Which device acts as an I/O device?
 - a) **Smartboard**
 - b) Keyboard
 - c) Whiteboard

Activity 1.3

Complete the sentences below.

1. The purpose of an interface in a machine is _____.
 - a) to control the machine's operations
 - b) **to provide functionality and usability**
 - c) to connect the machine to the internet

2. For a microwave oven, an embedded computer adds the functionality of _____.
 - a) cooking food
 - b) heating beverages
 - c) **programmable cooking modes**

3. For an air conditioner, an embedded computer adds the functionality of _____.
 - a) **automatically adjusting room temperature**
 - b) blowing hot air to warm up a room
 - c) blowing cold air to cool down a room

4. An example of a modern human-machine interface is _____.
 - a) dials and knobs
 - b) switches and buttons
 - c) **graphical user interface**

5. The first to have an embedded computer was the _____.
 - a) **Nokia 9000 Communicator**
 - b) Motorola DynaTAC 8000X
 - c) Apple iPhone

Activity 1.4

Answer the questions, and complete the sentences below.

1. Which of the following is an example of a task that a robot can perform in a factory?
 - a) Conducting underwater inspections
 - b) Exploring hostile environments
 - c) **Tightening screws on an assembly line**

2. Which category of robot can make decisions on its own without human intervention?
 - a) Semi-autonomous robot
 - b) Non-autonomous robot
 - c) **Autonomous robot**

3. A robotic vacuum cleaner that does not require human control is best described as _____.
 - a) stationary and non-autonomous
 - b) stationary and autonomous
 - c) **mobile and autonomous**

4. A robotic arm on an assembly line that requires some human control is best described as _____.
 - a) stationary and non-autonomous
 - b) **stationary and semi-autonomous**
 - c) mobile and autonomous

5. What is one potential benefit of using robots in hazardous environments?
 - a) Improved job opportunity
 - b) **Improved safety**
 - c) Improved cost savings

Activity 1.5

Answer the questions below.

1. Which types of computers do you use? List the devices that constitute each computer's interface.

Desktop: screen, keyboard, mouse, speaker, microphone, camera

Laptop: screen, keyboard, touchpad, speaker, microphone, camera

Tablet / smartphone: touchscreen, speaker, microphone, camera

2. What peripheral devices have you added to your computer's interface?

Wireless / wired earphone +\- microphone, webcam, printer, scanner

3. Of the peripheral devices you use, what functionality does each device provide you? How do you find that beneficial?

The printer allows me to print homework from the comfort of my home.

Earphones allow me to listen at higher volumes without bothering others.

Activity 1.6

Write your answer(s) on the lines provided below.

1. List smart machines found at home or school.

Smart (interactive) board

Smart thermostat or air conditioning unit

Smart TV

2. Most modern elevators have embedded computers that perform various tasks to ensure the safe and smooth operation of the elevator. Describe the user interface of a modern elevator.

Outside the elevator, there is a screen to show the floor number at which the cabin is currently at, as well as call buttons to call for the cabin.

Inside the elevator, there is also a screen to indicate the current floor, a control panel of floor numbers and a speaker to sound arrival at the requested floor.

3. “A smartphone is a computer, not a cellphone with an embedded computer.” Is this statement true? Explain.

This statement is mostly true as it depends on the main use or purpose of the phone. In general, the smartphone is mainly used for its general-purpose capabilities: navigating the web and social media, communicating via text or audiovisual messages, reviewing and completing assignments, attending online classes, playing games, listening to music, watching videos, etc.

Activity 1.7

Write your answer(s) on the lines provided below.

1. List three tasks robotic arms perform in factories.

Load and unload items

Weld and solder

Insert and tighten screws

2. Investigate the rovers sent to explore the surface of Mars. Choose one and list three of the various functions it can perform based on its sensors and actuators.

Curiosity landed on Mars in August 2012.

It has six wheel actuators that can move across rocky terrain.

It has sensors to monitor temperature, pressure, wind speed and radiation.

It has a robotic arm actuator to collect samples.

3. “Cars that can drive themselves are robots.” Is this statement true? Explain. .

If we define robots as machines capable of performing tasks autonomously

or under varying degrees of control, then self-driving cars can be considered

robots. These cars have sensors, actuators and onboard computers to perceive

the environment, make decisions and control their movements. They can

operate without direct human intervention.

